

Research report

Distinct brain activation patterns for human maximal voluntary eccentric and concentric muscle actions

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Abstract

Eccentric muscle contractions generate greater force at a lower level of activation and subject muscles to more severe damage than do concentric actions. A recent investigation has revealed that electroencephalogram (EEG)-derived movement-related cortical potential (MRCP) is greater and occurs earlier for controlling human eccentric than concentric *submaximal* muscle contractions. However, whether the central nervous system (CNS) control signals for high-intensity or *maximal-effort* eccentric movements differ from those for concentric actions is unknown. The purpose of this study was to determine whether the MRCP signals differ between the two types of *maximal-effort* contractions. Eight volunteers performed 40 maximal voluntary eccentric and 40 maximal voluntary concentric elbow flexor contractions on a Kin-Com isokinetic dynamometer. Scalp EEG signals (62 channels) were measured along with force, joint angle, and electromyographic (EMG) signals of the performing muscles. MRCP-based two-dimensional brain maps were created to illustrate spatial and temporal distributions of the MRCP signals. Although the level of elbow flexor muscle activity was lower during eccentric than concentric movements, MRCP-indicated cortical activation was greater both in amplitude and area dimension for the eccentric task. Detailed comparisons of individual electrode signals suggested that eccentric movements needed a significantly longer time for *early preparation* and a significantly greater magnitude of cortical activity for *later movement execution*. The extra preparation time and higher amplitude of activation may reflect CNS activities that account for the higher risk of injury, higher degree of movement difficulty, and unique motor unit activation pattern associated with maximal-level eccentric muscle actions.

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1. Introduction

All of our limb and body movements consist of concentric (shortening) and eccentric (lengthening) muscle contrac-

tions. Compared with concentric muscle actions, eccentric contractions exhibit many unique characteristics, including lower muscle activation for a same level of force at a given velocity [7,8,31,46], depressed corticospinal neuron [1,41] and monosynaptic reflex excitability [1,39], reduced ability to fully activate the working muscle [40,48], altered motor unit recruitment strategy from the “size principle” [25,28,32], reduced smoothness of the movement [19,36], and greater susceptibility to muscle tissue damage [34,42,47].

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Because most movement cycles consist of both eccentric and concentric muscle actions (e.g., during jumping, an eccentric movement generally occurs before the concentric action) and a large proportion of our daily living activities require accurate control of eccentric movements (e.g., returning a water glass onto the table surface or lowering a grocery bag into the vehicle trunk), eccentric movements are increasingly employed in medical rehabilitation and athletic training programs [9,10,23,27,37]. These training programs take physiologic advantages of characteristics associated with eccentric muscle activities, such as greater tension and power [4,15,35], higher energy efficiency [3,13], and adaptive changes to resist future tissue damage [12,20,24].

Despite many differences between the two types of muscle actions and the preferential application of one over the other in a variety of clinical and athletic programs, little is known regarding the control mechanisms underlying eccentric and concentric muscle activities. A recent study [19] reported that although the muscle activation level was significantly lower, the surface electroencephalogram (EEG)-derived movement-related cortical potential (MRCP) was greater and occurred earlier for eccentric than concentric contractions of elbow flexor muscles. These data suggest, for the first time, that the central nervous system (CNS) possesses unique strategies for controlling eccentric muscle actions [17,32].

Participants in that study [19] performed the two motor tasks against a load equal to 10% body weight. Greater discrepancies (such as tissue damage) between eccentric and concentric movements, however, occur when the exerted forces were high, and comparisons of maximal force, EMG, and ability to fully activate the motor neuron pool can only be made under the condition of maximal voluntary contractions (MVC). For these reasons, it is important to understand CNS control mechanisms for high-intensity eccentric and concentric muscle contractions. In addition, the study by Fang et al. [19] was limited to only four electrodes to record EEG signals from the scalp. Thus, it was not possible to detect how cortical electrical activities associated with the motor actions were distributed on the entire scalp surface with only four channels of EEG data. The purpose of this study was to determine spatial and temporal distributions of MRCP signals (based on 62 EEG electrodes) during planning and execution phases of MVC tasks of eccentric and concentric contractions of elbow flexor muscles. Preliminary results have appeared in abstract form [18].

2. Methods

2.1. Subject

Eight right-handed men (29.13 ± 2.36 years old, range 26–31 years) participated in the study. All individuals

were healthy and had no known neuromuscular disorders. The work was approved by the local ethics committee. All subjects gave informed consent prior to their participation.

2.2. Data recording

2.2.1. Mechanical recording

Subjects were seated comfortably in an experimental chair beside a Kin-Com isokinetic dynamometer (Chatteex, Chattanooga, TN). The subject's left arm was secured on the lever arm of the dynamometer and was kept at shoulder height. The upper arm was rotated 30° forward from a straight line connecting the left and right shoulders. The forearm was positioned in a neutral position between supination and pronation. The subject's shoulders, torso, and left upper arm were stabilized so that only movements at the elbow joint of the left arm were allowed (Fig. 1). The dynamometer could be programmed so that its lever arm could be rotated along its axis at the specified range, direction, and speed. An attached load cell on the lever arm measured the force exerted by the subject against the lever arm.

Subjects performed maximum voluntary concentric contractions by exerting elbow flexion force against the lever arm, rotating it toward the body (the elbow flexors were shortening). They performed maximum voluntary eccentric contractions of the elbow flexor muscles against the lever arm, while lever arm rotated away from the body (the elbow flexors were lengthening). The range of lever arm or elbow joint rotation was 30° for each concentric or eccentric contraction. The angle signal was recorded both in the Spike2 (for peripheral data) and NeuroScan (for EEG data) systems (see below) to synchronize the EMG and EEG signals. A trigger signal was generated and stored into the computers for the subsequent triggered-averaging of the EEG and peripheral signals (EMG, force, and position)

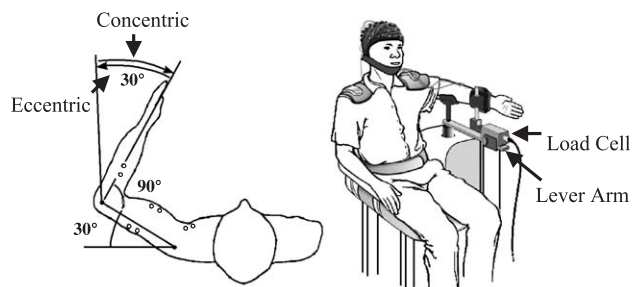


Fig. 1. An illustration of the experimental setup. The left panel shows a top view of the angle and range of movements. As indicated by the two thin arrows in the left panel, the eccentric contraction involved the forearm moved away from the midline of the body (muscle lengthening), which was a result of a greater external force than the force produced by the muscles and the concentric contraction involved the forearm moved towards the midline of the body (muscle shortening), in which the muscle force was greater than the external resisting force.

when the rotation (changing of angle) reached a 3° threshold in each trial. The force and position outputs were directed to a laptop computer equipped with data acquisition hardware and software (Spike2, Cambridge Electronic Design, Cambridge, UK). The force and position (angle) data were digitized at a rate of 100 samples/s and saved on the hard disk of the laptop computer.

2.2.2. EMG recording

Surface EMG signals were simultaneously recorded from the biceps brachii (BB), brachioradialis (BR), triceps brachii (TB), and deltoid (DT) muscles. The BB and BR are elbow flexors and the TB is the extensor. The DT muscle abducts the arm at the shoulder joint (electrodes were placed on the middle portion of the DT). Bipolar electrodes were attached to the skin overlying the belly of each muscle. Inter-electrode distance was kept at about 2 cm. The reference electrode was fixed on the skin overlying the lateral epicondyle near the elbow joint. The EMG signals were amplified ($\times 1000$) and filtered (10–3000 Hz) using Grass Neurodata amplifier systems (Astro-Med, West Warwick, RI). The output signal from the EMG amplifiers was digitized (1000 samples/s) using the Spike2 system and saved on the hard disk of the laptop computer.

2.2.3. EEG recording

A 64-channel NeuroScan EEG system (NeuroScan Labs, Sterling, VA) was used to acquire EEG signals from the scalp. A Quik-Cap elastic nylon cap that holds 64 surface electrodes (Neuromedical Supplies, Sterling, VA) was placed on the scalp for EEG data recording. The configuration of the electrode arrangement in the cap is based on the International 10-20 System [26]. Recording diameter was 6 mm and cavity depth was 5 mm for each electrode. Conducting gel (Electro-gel™, Electro-Cap International, Eaton, OH) was injected into each electrode to connect the recording surface of the electrode with the scalp. An impedance map was displayed on a computer monitor while the investigators filled the electrodes with the gel. In general, data recording did not begin until the impedance for all electrodes settled below 10,000 Ω . The vertical electro-oculogram (VEOG) electrode was used to record angle information, which was used for the triggering purpose. The HEOG (horizontal electro-oculogram) electrode was not used. The other 62 active electrodes were referenced to the common linked earlobes. All channels of the EEG signals were amplified ($\times 75,000$ [headbox $\times 150$, main amplifier X500]), low-pass filtered (0–50 Hz), and digitized (250 sample/s) using the NeuroScan Labs software.

2.3. Experimental Procedures

The EEG electrode cap was first loosely fixed on the head and the earlobe electrodes were attached after scalp

skin preparation. The cap was then pulled down firmly over the head, with the midline frontal pole electrode (FPZ) being located approximately 4 cm above the nasion. Three different sized caps (small, medium, and large) were available to best fit the subject's head. Conducting gel was injected into each electrode using a syringe. The color map indicating impedance of the electrodes was displayed for online improvements of the connection between the electrodes and scalp. High impedance was usually reduced by injecting more gel or applying pressure to the electrode. A nylon net that covered the entire electrode cap was placed on the cap and pulled down to be tightened under the subject's chin. This net applied additional pressure to the electrodes. After completion of the electrode application, subjects sat quietly for ~ 5 min. During this time, the impedance usually drifted down to a stable value ($< 10,000 \Omega$) for those electrodes initially showing high impedance values, and EMG electrodes were attached to the skin overlying the respective muscles (BB, BR, TB, and DT). The entire preparation took about 20 min.

Each subject performed 40 concentric and 40 eccentric maximum voluntary elbow flexor muscle contractions of the left arm. The contractions were performed alternatively, i.e., concentric, eccentric, concentric, eccentric, and so on, and were self-paced, with a ~ 20 -s rest between each two adjacent trials. Alternative performance of concentric and eccentric contractions balanced possible fatigue effects on brain and muscle activities for the two tasks. (To determine if fatigue had occurred, the mean value of the last three MVC forces was compared with that of the first three for both eccentric and concentric tasks.) The range and velocity of the lever arm rotation were $30^\circ/\text{s}$ and $30^\circ/\text{s}$, respectively. During each concentric contraction, the elbow joint rotated from the 120° position to the 90° position (180° equaled the elbow being fully extended). During each eccentric contraction, the elbow joint rotated from the 90° position to the 120° position (Fig. 1). Subjects were told to avoid eye blinks, biting the teeth, or tensing the facial and neck muscles in a period from ~ 5 s before to the completion of each movement, as these activities create unwanted noise in the EEG data. However, these activities were allowed during the initial 15 s of each 20-s rest period.

2.4. Analysis

All raw EEG data were inspected visually. Trials that contained eye blinks or other signal artifacts were excluded. These artifact signals are usually of short duration but of substantially higher amplitude and frequency than normal EEG signals and, therefore, are relatively easy to recognize. The baseline EEG data were pre-corrected according to the mean value of -4200 to -2200 ms recordings, during which subjects sat quietly and cortical preparation for the movement had not begun. For each contraction, the trigger

signal triggered a 6-s window of the EEG recording, 4 s before the trigger and 2 s after the trigger. The force, angle, EMG (EMG data was rectified first), and EEG data were trigger-averaged across the “clean” trials (ranged from 35 to 40 trials among the subject) using the same trigger signals. The mean values of force and EMG were then calculated based on the data points located between the starting and ending points of the movement (indicated by the angle signal). Rate of force development was measured by determining the slope of the force rising curve from the averaged force record in each subject. Previous reports have shown greater MRCP is associated with a higher rate of force development [44]. MRCP of each electrode in each subject was obtained after averaging the EEG data. Finally, a grand average of the MRCP of each channel across the eight subjects was performed.

2.4.1. Two-dimensional (2-D) mapping of MRCP

The mean MRCP of the eight subjects was used for 2-D mapping of electrical potential of the brain, which drew topographical maps of MRCP showing spatial and temporal distributions of cortical electrical activities during the planning and execution phases of each of the two tasks. The maps were based on the MRCP data of 62 EEG electrodes. The analysis included 1960 ms, beginning from 2120 ms (–2120 ms) before the trigger to 160 ms (–160 ms) before the trigger. (On average, EMG of the BB began at 730 ms, force at 648 ms, and movement at 160 ms before the trigger.) Each map in each task displayed a spatial distribution of a mean potential of 40 ms. The MRCP or negative potential (NP) onset time was calculated based on the group average of NP. As soon as the NP value of any of the 62 electrodes reached the $-3 \mu\text{V}$ threshold (see below), that time point (based on 40-ms averaging) was taken as the NP onset time for the task. Because the potential maps were based on mean NP values of the entire group, they are more likely to reflect true cortical activities associated with controlling the motor tasks rather than representing a random or selective case based only on single-subject data.

To quantify the area of brain activation based on the NP group average, the number of electrodes whose amplitude was higher than the threshold was counted. The threshold of activation was determined by standard deviation (S.D.) of the baseline potential of the group average data. The baseline potential was calculated from –4200 to –2200 ms. The mean value of the baseline was zero after baseline correction. For the eccentric task, there were 41 electrodes in the group average data whose baseline SD was less than $-1 \mu\text{V}$, 56 electrodes less than $-2 \mu\text{V}$, and all 62 electrodes less than $-3 \mu\text{V}$. For the concentric task, the baseline S.D. in 37 electrodes was less than $-1 \mu\text{V}$, in 61 electrodes less than $-2 \mu\text{V}$, and in all 62 electrodes less than $-3 \mu\text{V}$. Since the baseline S.D. of all electrodes was less than $-3 \mu\text{V}$, this value ($-3 \mu\text{V}$) was selected as the threshold to determine whether the potential for a particular electrode was elevated. The potential of each electrode was calculated for every 200-ms period beginning

from –2120 to –120 ms, with the mean amplitude greater than $-3 \mu\text{V}$ over the 200-ms period being considered as a significant increase in the potential. The number of electrodes meeting this criterion at 10 time points (–1920, –1720, –1520, –1320, –1120, –920, –720, –520, –320, and –120 ms) was counted to determine activation area at these time points. We chose 200-ms averaging instead of 40 ms for each data point to avoid a crowded data plot (10 vs. 50 data points).

Electrodes C3 (over the ipsilateral [left hemisphere] primary motor area), C4, C6 (over the contralateral [right hemisphere] motor areas), and F4, FC4, and FC6 (over the contralateral regions of the frontal areas anterior to electrode C6) were selected for detailed MRCP value analysis. These six electrodes cover major motor areas in the frontal lobe and showed strong MRCP.

MRCP NP was measured in each of the six electrodes. The MRCP consists of NP and positive potential. NP occurs before onset of movement and thus is in the open-loop portion of the movement without a significant influence from the feedback signals. Generally, the NP is thought to represent cortical activities associated with planning and execution of the motor activity. The NP was further divided into readiness potential (RP) and negative slope (NS). Because the RP occurs before the NS, RP is considered to represent cortical activities that are related to early preparation for a muscle action and NS is more associated with specific planning and execution of the motor activity [14,44]. In many cases, because it was difficult to clearly identify a single point indicating the beginning of the RP and NS, the RP and NS were measured using a curve-fitting method: a straight line was drawn along the approximate shape of the baseline potential, the RP, and the NS [19,44]; the intersection of the two lines was taken as the beginning of the subsequent potential. For example, the intercept of the baseline and RP line was the onset of the RP and the intercept between the RP and NS lines was the onset of NS (Fig. 2).

Two values, amplitude and time latency, were determined for the RP, NS, and NP for each of the six electrodes. For the RP, the amplitude was measured from the onset of RP to the onset of the NS, and the latency was calculated from the onset of RP to the onset of the EMG (BB). For the NS, the amplitude was measured from the onset of the NS to the peak of the NP, and the latency was determined from the onset of NS to the onset of the EMG (BB). The NP amplitude was the sum of the amplitude of the RP and NS, and the NP latency was the same as the RP latency. The onset of EMG was determined by taking the intersection of the baseline and the straight line drawn along the rising slope of the EMG. The RP and NS values of the six electrodes of each subject were then used for the group averaging.

2.4.2. Force fluctuation

Greater force fluctuation (tremor) was reported for submaximal eccentric than concentric tasks when static

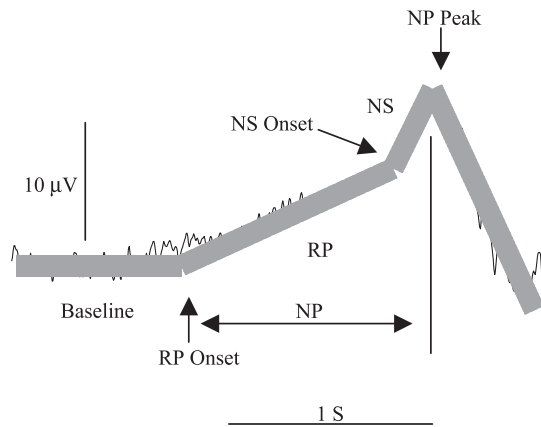


Fig. 2. Components of movement-related cortical potential (MRCP) and illustrations of their measurements. RP, readiness potential; NS, negative slope; NP, negative potential. RP amplitude was measured from baseline to NS onset and latency from RP onset to EMG (biceps brachii) onset (not shown). NS amplitude was determined from NS onset to NP peak and latency from NS onset to the EMG onset. NP amplitude was the sum of the amplitude values of RP and NS. NP latency was the same as RP latency.

force was maintained [19]. In this study, force fluctuation was assessed from the first and last three trials of the concentric and the eccentric contractions. In each trial, the S.D. of the mean force was calculated during the movement range. Determining the S.D. from the beginning and end trials of each task would have balanced any learning or fatigue effect on the measurement (S.D.).

2.5. Statistical analysis

Paired *t*-tests were performed to compare force, EMG, and measurements of MRCP between concentric and eccentric muscle contractions. Because we expected that each measurement for the eccentric task would differ towards a particular direction from the corresponding measure for the concentric task [19], one-tailed *t*-tests were employed for the comparisons. Considering the multi-channel-comparison nature of the analysis, we applied additional analyses using Hotelling *T*-Square test for the multi-channel eccentric to concentric comparisons. Hotelling *T*-Square test is a statistical method for pair-wise multivariate statistical analysis [2] while maintaining the same statistical power. Significance level was determined at $P \leq 0.05$, and the data were reported as mean $\pm$ S.D. unless otherwise mentioned.

3. Results

3.1. Force and EMG

The force exerted by the subjects during eccentric MVCs (210.63 ± 94.68 N) was significantly higher ($P < 0.05$, Figs. 3 and 4A) than the force during concentric MVCs (184.06 ± 87.07 N). The force results agreed with results

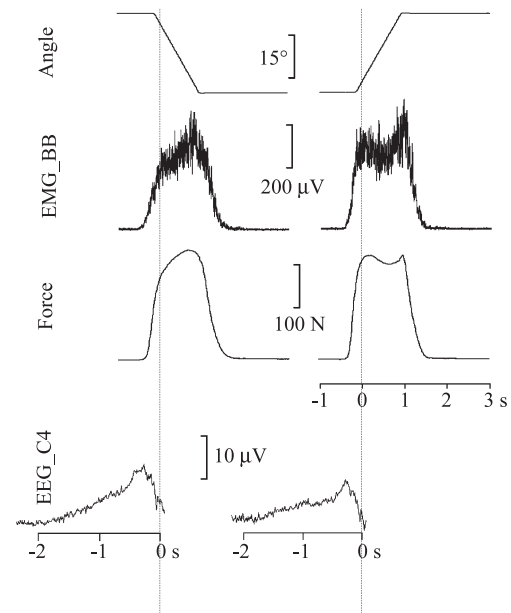


Fig. 3. Examples of joint angle, EMG (biceps brachii), force, and MRCP (C4 electrode) for eccentric (left) and concentric (right) contractions of a subject. Long vertical lines indicate the timing of trigger. Note that in this example although the muscle activity level (EMG amplitude) was lower, the MRCP was greater for eccentric than concentric muscle action.

from most studies in the literature that also reported greater eccentric force (e.g., Refs. [15,35]). The rate of force development was similar ($P > 0.5$) between the eccentric

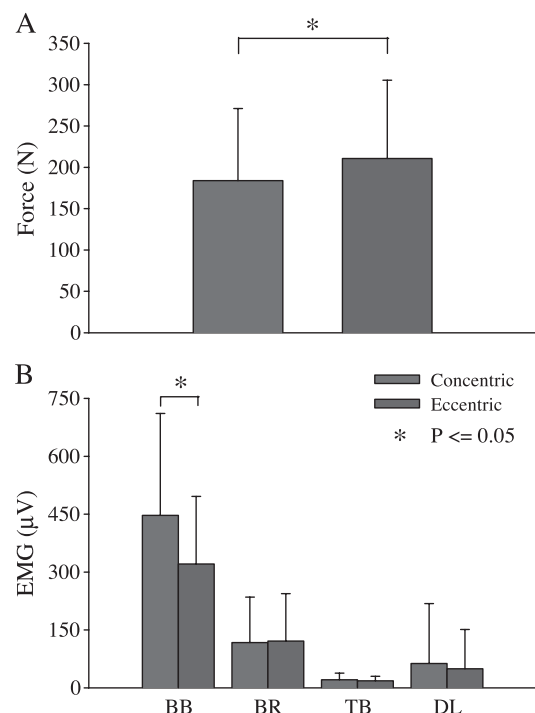


Fig. 4. Force (A) and EMG (B) measured during eccentric and concentric contractions of the elbow flexor muscles. The force for the eccentric task was significantly higher but the EMG for the task was significantly lower than the corresponding values of the concentric task. BB, biceps brachii; BR, brachioradialis; TB, triceps brachii; DL, deltoid muscles.

(166.64 ± 78.38 N/s) and concentric (193.13 ± 86.29 N/s) tasks.

The EMG of a major elbow flexor muscle (BB) during eccentric contraction (321 ± 175 μ V) was significantly lower ($P < 0.01$, Figs. 3 and 4B) than the EMG of the muscle during concentric contraction (447 ± 264 μ V). This result is consistent with most other studies that investigated EMG values between the two types of muscle activities (e.g., Refs. [7,19,46]). The EMG data for the BR (another elbow flexor) was not significantly different (Fig. 4B) for the two tasks (121 ± 123 μ V for eccentric and 117 ± 118 μ V for concentric). This result was probably related to the forearm position at which the tasks were performed. Normally when the arm is positioned on a sagittal plane (e.g., elbow resting on a support surface and the forearm is moved upward towards the shoulder by contracting the elbow flexor muscles), the BR is optimally activated if the forearm is in the neutral position. However, in our experiment, the arm was held on the horizontal plane and the subject attempted to contract the muscles to move the forearm towards the midline of the body (see Fig. 1). Under this condition, the BR was not optimally activated when the forearm was in the

neutral position. The EMG values for the TB (18 ± 12 μ V for eccentric and 21 ± 17 μ V for concentric) and DT (49 ± 102 μ V for eccentric and 63 ± 155 μ V for concentric) muscles were similar ($P > 0.1$, Fig. 4B).

The observation of higher eccentric force with lower muscle activities is consistent with the findings of former studies [7,15,31,35], and is attributed to the additional force contributed by the stretched elastic components in series and/or parallel to the contractile elements.

3.2. Two-dimensional MRCP mapping

Group averages of MRCP were obtained, and the MRCP waveforms for all 62 channels were topographically displayed (Fig. 5). A majority of electrodes showed a greater NP for eccentric than concentric movements. Those that did not demonstrate a difference in NP were located far away from sensorimotor-related areas (e.g., the occipital lobe). In Fig. 5, the MRCP waveforms of a number of electrodes have been enlarged to illustrate the striking difference between the two tasks (blue for eccentric and red for concentric). Based on the grouped MRCP data, a series

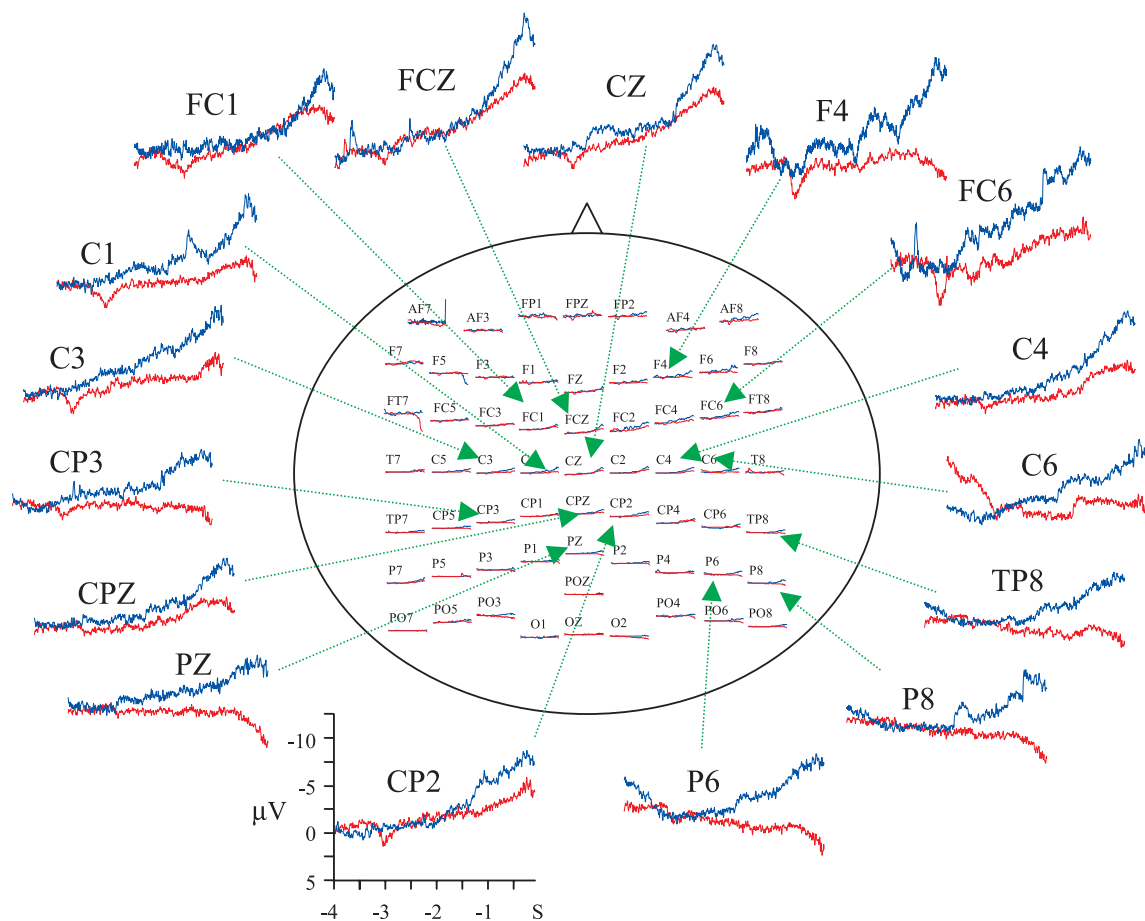


Fig. 5. A multi-channel display of MRCP for concentric (red) and eccentric (blue) muscle activities. The display was based on average data from 8 subjects. A number of channels have been selectively enlarged to more clearly demonstrate differences in the MRCP signal between the two types of muscle actions. Almost all channels that recorded an MRCP showed greater magnitude of the signal and earlier rising of the potential during eccentric than concentric movements. The triangle on the top of the circle indicates direction of the nose and the MRCP map is viewed from the top of the head.

of 2-D maps were created beginning from -2120 to -160 ms (Fig. 6).

The 2-D mapping of MRCP allows examinations of spatial and temporal distributions of cortical activities

during the two types of muscle activities. Several major differences occurred in both temporal and spatial distributions, which suggests that cortical activation patterns between the controlling processes of eccentric and concen-

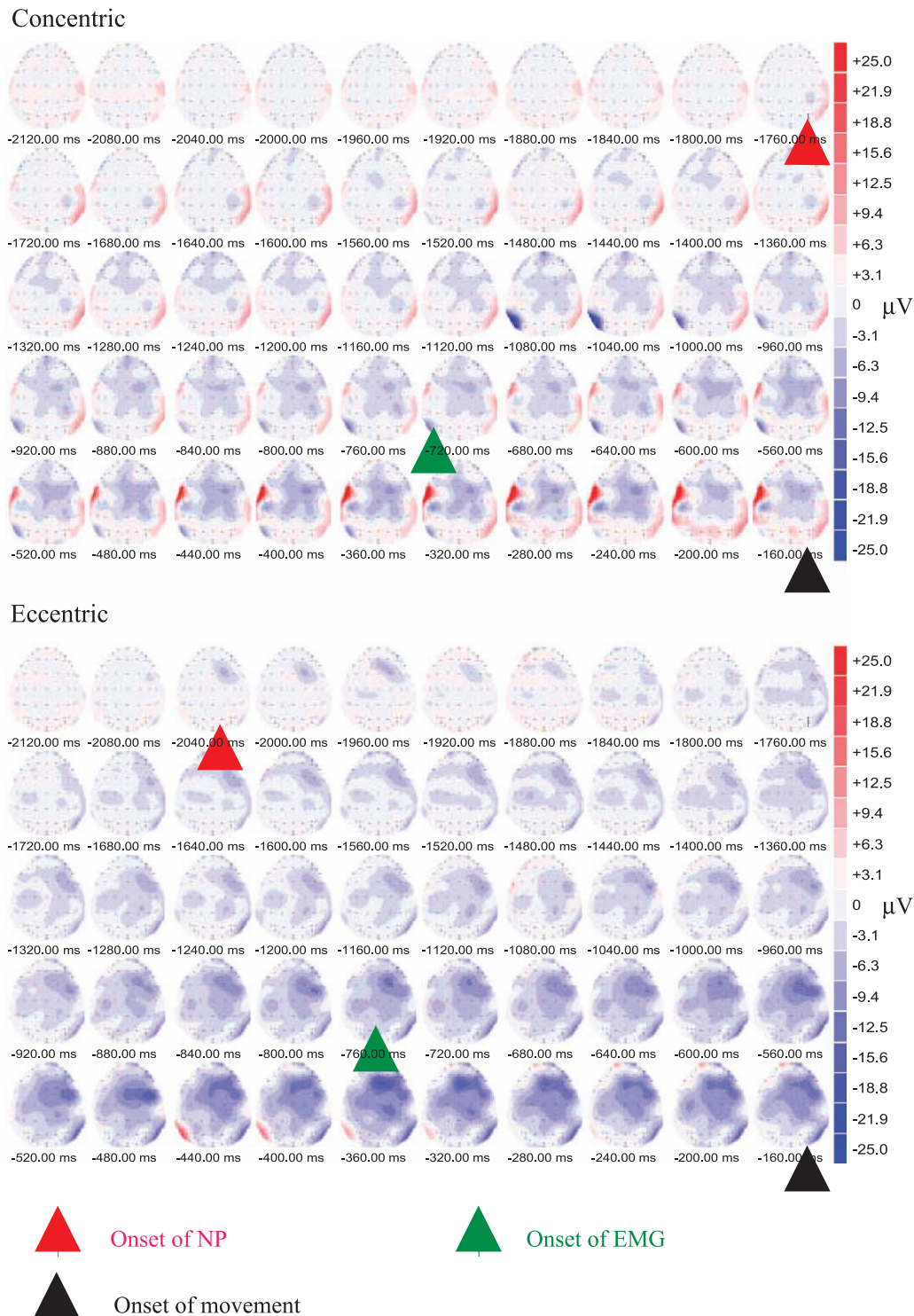


Fig. 6. Group average of topographical maps of cortical potential (μV) showing spatial and temporal distributions of electrical activity at the cortical surface for concentric (upper panel) and eccentric (lower panel) elbow flexor MVC contractions. The maps show that (1) the preparation was longer and started earlier for eccentric than concentric contractions, (2) the negative potential was greater during eccentric than concentric contractions, especially during late preparation and execution periods in the frontal and parietal lobes, and (3) the activation area was larger for eccentric than concentric movements.

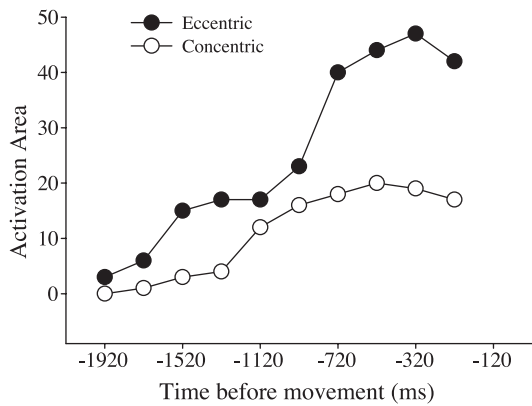


Fig. 7. Activation area determined by number of electrodes passing a threshold ($-3 \mu\text{V}$) during eccentric (filled circles) and concentric (open circles) muscle activities. A larger activation area was detected at all time points. As the time got closer to movement execution, the difference in activation area between the two tasks became more prominent.

tronic movements are not the same (Fig. 6). First, the onset of NP for the concentric task began about 1760 ms before the trigger but that for the eccentric movement started 2040 ms before the trigger, a 280-ms difference. Second, the electrodes showing onset activities for the eccentric task were located at the FC4 and FC2 positions in the right frontal area (movements occurred on the left side), whereas for the concentric task, onset activity first occurred at the CP4 location in the right central-parietal area (Fig. 6, red arrows). Third, the NP is distributed in a substantially larger area (see below) with higher amplitude for the eccentric than concentric tasks.

3.3. Brain activation area

To quantify cortical activation area for the two motor tasks, the number of electrodes that passed the NP threshold ($-3 \mu\text{V}$) was counted from 2120 to 120 ms (every 200 ms) before trigger. For the eccentric task, the number of electrodes with NP greater than $-3 \mu\text{V}$ at each of the 10 time points was 3, 6, 15, 17, 17, 23, 40, 44, 47, and 42, respectively. The numbers of electrodes showing the NP greater than $-3 \mu\text{V}$ at the 10 time points were 0, 1, 3, 4, 12, 16, 18, 20, 19, and 17, respectively, for the concentric contraction (Fig. 7).

3.4. Detailed MRCP analyses for electrodes C3, C4, C6, F4, FC4, and FC6

Detailed values of the RP and NS amplitude and latency of the six selected electrodes are given in Table 1. In general, compared to the concentric task, the RP amplitude did not change but its latency prolonged significantly during eccentric muscle contractions. In contrast, the NS amplitude during the eccentric action elevated significantly compared to the concentric task but the latency of the NS did not change in almost all of the six electrodes (Figs. 8 and 9).

Table 1
Measurements of amplitude (μV) and the latency (ms) of the readiness potential (RP) and negative slope (NP) for the eccentric and concentric tasks

	Eccentric						Concentric						Hotelling's
	C3	C4	C6	F4	FC4	FC6	C3	C4	C6	F4	FC4	FC6	
RP amplitude	3.72±2.66	5.62±2.31	3.12±1.48	4.22±4.21	4.81±4.51	4.55±2.51	1.74±2.84	3.12±2.84	2.10±1.63	1.14±3.03	4.58±3.79	2.89±1.74	$P>0.5$
One-tailed <i>t</i> -test*	$P>0.1$	$P<0.01$	$P>0.2$	$P<0.02$	$P>0.4$	$P>0.1$	1.75±3.99	3.08±2.13	0.07±3.52	-0.39±3.71	2.84±4.99	0.33±4.94	$P<0.01$
NS amplitude	3.74±3.68	5.27±2.09	3.84±2.71	6.08±6.05	5.95±6.54	5.55±4.17	1027±462	994±473	1079±462	1035±270	762±384	833±418	$P<0.05$
One-tailed <i>t</i> -test	$P<0.005$	$P<0.02$	$P<0.02$	$P<0.01$	$P<0.05$	$P<0.01$	51±267	55±359	212±248	11±336	8±185	-20±214	$P>0.4$
RP latency	1355±334	1388±361	1273±334	1357±235	1126±381	1187±348	105±216	105±216	105±216	105±216	105±216	105±216	
One-tailed <i>t</i> -test	$P<0.02$	$P<0.01$	$P<0.04$	$P<0.001$	$P<0.01$	$P<0.003$							
NS latency	150±293	60±242	221±222	65±240	19±278	105±216							
One-tailed <i>t</i> -test	$P>0.3$	$P>0.4$	$P>0.4$	$P>0.3$	$P>0.4$	$P<0.03$							

* One-tailed *t*-test comparing the values of eccentric and concentric tasks of a given electrode.

§ Hotelling *T*-Square test for comparisons of the six-channel data between the eccentric and concentric tasks.

3.5. Force fluctuation

The S.D. of the mean force value was measured for the first and last three trials for both eccentric and concentric tasks. A mean value was obtained by averaging the S.D. values of the six trials for each task for each subject before a group average for each task was determined. Mean force fluctuation was 34.15 ± 19.57 N for eccentric action and 21.80 ± 14.82 N for concentric movement ($P < 0.01$).

3.6. Fatigue effect

The mean peak force for the first three eccentric contractions was 185.45 ± 81.27 N and 180.09 ± 118.84 N for the last three trials ($P > 0.5$). The mean peak force for the first three concentric contractions was 145.22 ± 91.96 N, and 157.25 ± 105.85 N for the last three trials ($P > 0.5$). The similarity in the maximal force measured at the beginning and end of the experiment for each task indicates that subjects were not significantly fatigued by performing the total number of muscle contractions.

4. Discussion

The purpose of this study was to determine whether EEG-measured brain activities (amplitude, timing, and distribution) associated with maximal eccentric voluntary contractions of human elbow flexor muscles differ from those related to concentric ones. The two major findings were: (1) MRCP NP values, which are related to movement preparation and execution, were greater and more widely distributed on the scalp during eccentric than concentric tasks; the increase of NP amplitude occurred mainly in the NS primarily contributed by cortical activities for movement execution. (2) The onset times of the NP were earlier for eccentric than concentric muscle contractions, which particularly occurred in RP in general preparation for the coming movement.

4.1. MRCP NP

The NP consists of RP and NS. The RP is related to general preparation for the motor task [14,16,22,29,30], and it probably represents cortical activities associated with attention, concentration, and early preparation for the movement. The NS, which occurs right before the movement and begins with a steeper rising slope, is thought to be directly related to cortical activities of specific planning, programming, and execution of the motor task [5,22,33,43,44]. In this study, we found that only the NS showed convincing amplitude increases for the eccentric task as compared with those for concentric movement. Although the eccentric RP showed a trend toward higher amplitude (Fig. 8A), its increase over the concentric RP was not as prominent, with large variation among the subjects. These data indicate that the greatest increase in eccentric

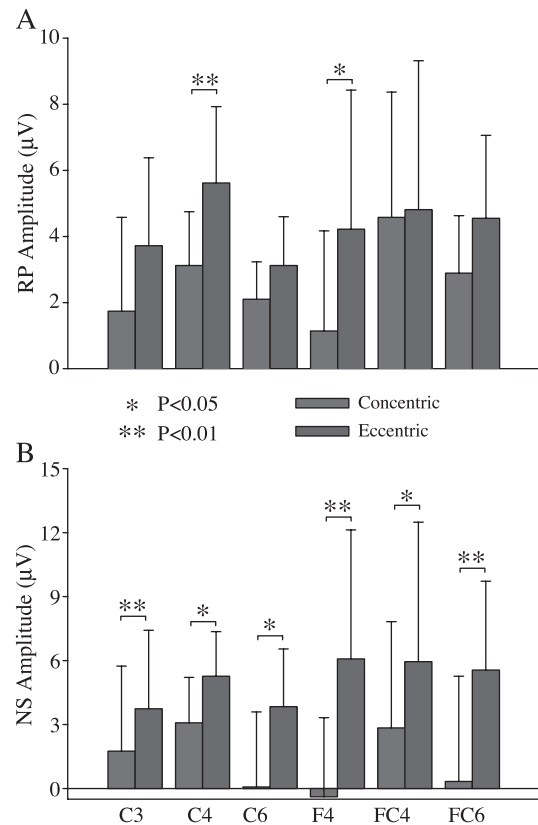


Fig. 8. Amplitude of readiness potential (RP) (A) and negative slope (NS) (B) measured from 6 electrode locations for eccentric and concentric muscle contractions. Although there is a trend for RP amplitude to become higher for eccentric than concentric tasks, the value from four out of six electrodes did not reach statistical significance. On the contrary, the NS magnitude of all six electrodes was greater during eccentric than concentric muscle actions.

NP over concentric NP occurs at the later stage, beginning from about the onset of NS (Fig. 8B). The data imply that the level of cortical activation was substantially higher for eccentric than concentric tasks during the specific planning and execution phases of the controlling process.

The latency data (Fig. 9) suggested a longer preparation time for an eccentric than a concentric movement. However, the prolonged preparation mainly occurred during the RP stage. The latency from the NS to the initiation of eccentric muscle activity was not significantly longer than the corresponding latency for a concentric movement. The data imply that the CNS does not need additional time for execution of a maximal eccentric contraction. Additional time, however, is needed in the early preparation for the eccentric task. The following sections discuss possible explanations for the greater NP and longer preparation time needed to perform a maximal eccentric muscle contraction.

4.2. Potential mechanisms contributing to greater NP amplitude and latency

4.2.1. Tissue damage

It is well known that eccentric muscle contractions induce greater tissue damage than do concentric muscle

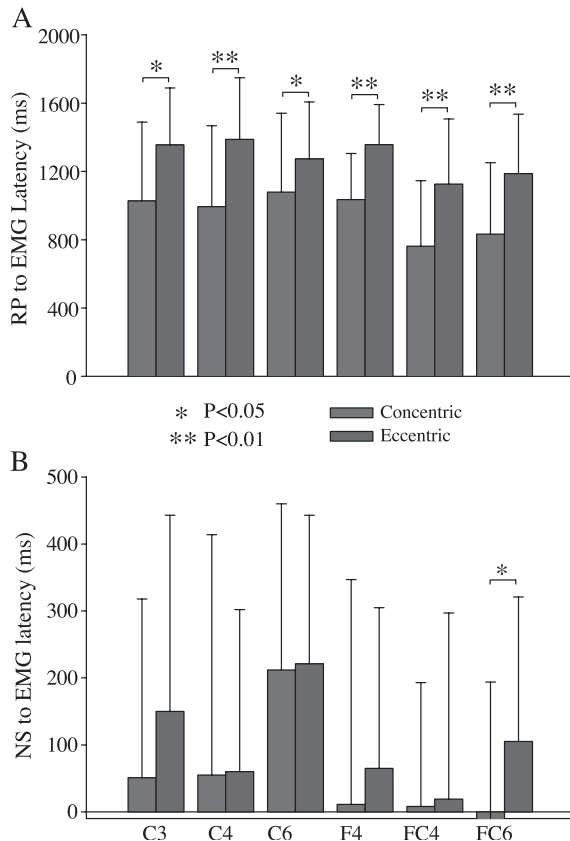


Fig. 9. Latency of RP (A) and NS (B) measured from 6 electrode locations for eccentric and concentric muscle contractions. The latency was measured from RP or NS onset to the onset of biceps brachii EMG. The RP latency was significantly longer for eccentric than concentric muscle activities at all six selected electrodes. The difference in the NS latency between the two tasks was small, only one electrode (FC6) showing a significant difference.

activities [21]. This observation is attributed to the higher force associated with eccentric muscle actions, especially higher force that each fiber bears (because fewer muscle fibers are recruited during an eccentric action [48]). The widely reported adaptive changes following repetitive eccentric muscle activities or eccentric training may be an indication of CNS-modulated adaptations to protect muscle from further injury. When the CNS prepares an eccentric MVC with the expectation of substantially greater force to be generated, it may need special preparation to limit or suppress the level of muscle activation to lower the risk of injury. This hypothesis is consistent with the observation [48] that an isolated muscle complied with the force–velocity relationship very well during eccentric contractions; the force increased exponentially with the lengthening velocity. However, when the same experiments were performed in vivo, the force did not change significantly with the lengthening velocity from 30°/s to 270°/s. It seemed that further force increases associated with higher lengthening velocity observed in the in vitro preparation were inhibited in vivo. The potential “damage reduction” activities in the

cortex may contribute to the greater NP signals and longer preparation time.

A specific CNS strategy to reduce potential tissue damage could be gating the presynaptic Ia afferent input from the lengthening muscle to attenuate elicitation of the unwanted stretch reflex at the spinal cord level. Because the force is already high during an eccentric contraction, the non-attenuated stretch reflex would further increase the force, which would lead to greater tissue damage. This speculation is supported by convincing evidence that monosynaptic reflex excitability is depressed during eccentric contractions of human upper and lower extremity muscles [1,39]. The modulation of reflex activities by the CNS is likely to magnify the total control signal (amplitude, timing) and to be reflected by the MRCP measurements. Other reflex pathways, such as those through recurrent inhibition, Ib or Golgi tendon organ and other feedback circuits also play an important role in regulating excitability of the motoneuron pool and force output of the muscles. However, their role in modulating muscle output during eccentric contractions is less known.

4.2.2. Degree of difficulty

Eccentric movements are more difficult to control than concentric ones. This fact was evidenced by greater force fluctuation or movement variability during eccentric contractions observed in this and previous studies [11,19,36]. Greater voluntary effort is required when a person performs a motor task that is more difficult to control. The exact mechanisms underlying greater force fluctuation during eccentric movements are not clear, but they may be related to periodic stretch reflex-induced EMG bursts and selective recruitment of larger-size motor unit for eccentric contractions. Both mechanisms would contribute to greater force fluctuation. Stretch reflex does not occur in a concentric contraction because the muscle is shortening. During an eccentric contraction, the muscle is likely lengthened by an external force and the spindle receptors that sense muscle length are stimulated, which give rise to the alpha motoneurons projecting to the same muscle being stretched. The stretch reflex induces muscle activities in addition to the ongoing eccentric contraction. The added muscle activity would increase the force, which would oppose muscle lengthening. However, the external force would stretch the muscle and produce another cycle of stretch reflex.

It has been reported that larger motor units are selectively recruited during eccentric muscle contractions [25,28,32]. Selective recruitment of large motor units would increase force fluctuation because each discharge of a large motor unit would generate a greater twitch force than would the discharge of a smaller motor unit because a large motor unit innervates a greater number of muscle fibers. As a result of the stretch reflex and selective recruitment of large motor units, force fluctuation is greater during eccentric movements, which require a greater effort or cortical activity to

maintain a smooth eccentric contraction. Neuroimaging studies have shown that when motor tasks with a higher degree of difficulty are performed, the level of brain activation is higher [38,49].

4.2.3. Different control mechanism

It has been suggested that eccentric contractions have a different motor unit recruitment pattern than concentric contractions. **In contrast to the concentric contraction, during which low-threshold motor units are recruited first, high-threshold motor units are selectively recruited during eccentric contractions [25,32]. Consistent with these results is Fridén's report [21] that type II muscle fibers, which are fast-twitch fibers belonging to large, high-threshold motor units, are predominantly affected after eccentric contractions. A different motor unit recruitment pattern may reflect a unique nervous system control strategy for eccentric movements, and that strategy may need greater cortical activity to carry it out.** However, other studies [6,45] suggested that selective recruitment of high-threshold motor units does not occur during human wrist flexor eccentric contractions.

Studies [1,41] using transcranial magnetic stimulation (TMS) of the brain to examine excitability of corticospinal neurons also support the notion of different control strategies for eccentric and concentric muscle actions. These studies reported significant reductions in the excitability of the corticospinal output neurons during eccentric than concentric elbow flexor muscle contractions. The TMS-induced motor evoked potentials (MEP) from the elbow flexor muscles during the eccentric movement, on average, was more than 30% smaller than that during the concentric muscle action. These results suggest that the excitability of the corticospinal pathway is suppressed, perhaps by higher-order cortical centers, such as association cortices when an eccentric muscle action is executed. This suppression may be related to the “plan” of limiting the size of the output signal of the corticospinal system to contain the magnitude of the force. This finding is likely to be linked with our observation of increased amplitude of MRCP and previous reports of reductions in spinal motoneuron excitability during muscle eccentric movements. Because the corticospinal neuron excitability is suppressed in an eccentric action, a greater command signal (MRCP) is needed to activate the muscle. Furthermore, the depression of the corticospinal neuron excitability per se or the modulating mechanism that leads to the depression is likely to reduce excitability of the spinal motoneuron pool [1,39].

4.3. Comparisons to submaximal contractions

MRCP NP during eccentric elbow flexion contractions against 10% body weight was greater and onset time was earlier than these two values during the concentric movement [19]. Comparing the results of the current MVC-

intensity study (based on the six selected electrodes) with the previous submaximal-intensity study (based on the four available electrodes), the relative increase in NP amplitude for eccentric over concentric actions, on average, was greater ($P<0.05$; a $234\pm 80\%$ increase for the MVC study vs. a $130\pm 5\%$ increase for the 10% body weight study). The increase in the NP onset time (from the EMG onset) for the eccentric over the concentric task was also longer ($P<0.05$) for the MVC study ($135\pm 11\%$) compared to the increase for the submaximal-intensity study ($119\pm 5\%$). The data suggest that the higher the activation level of the muscles, the greater magnitude and longer preparation time are needed for controlling an eccentric movement.

Because the 64-channel system was not available at the time of the previous study, we were unable to create a spatial map of the MRCP during the two types of muscle activities. Thus, it is not clear whether the area of the brain activated was greater for the submaximal eccentric movement than for the submaximal concentric contraction. In addition, due to specific characteristics of the task (holding the load, then releasing it—eccentric; raising the load, then holding it—concentric) of the earlier submaximal-load study, separating the RP from the NS was difficult (compare Fig. 2B of the former study [19] to Fig. 2 of the current work). The present study demonstrated that in controlling maximal eccentric elbow flexion contractions, prolonged MRCP NP onset time mainly occurs during the early preparation phase and greater NP amplitude is particularly needed for executing the movement.

4.4. Summary

The current study, involving MVC intensity, confirmed the results of our previous investigation: when a subject is resisting a submaximal load, the brain needs more time to prepare and larger magnitude of activities to carry out eccentric muscle contractions. Significant new findings indicate that a longer time is needed for the early phase of preparation and greater signal amplitude is needed for executing an eccentric movement. Furthermore, not only is the intensity of cortical signals for controlling eccentric muscle actions higher (indicated by higher amplitude of MRCP), but also the area of the brain involved in the control process is larger, which may indicate an involvement of more functional regions and a larger number of neurons in the brain to control eccentric than concentric muscle activities. Whether these unique strategies hold in regulating the eccentric phase of a rapid cycle of lengthening (eccentric) and shortening (concentric) movements in motor tasks such running and jumping needs further investigation.

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